Olbrecht 2021 HGSOC — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 samples / 5 patients share ONE continuous manifold (single kNN component, no detached islands). Per-neighbourhood iLISI rises 1.00→3.33 under Harmony (θ=10): sample-integrated, not a per-patient archipelago. Judged on the regenerated embedding, not a scalar alone.

Gate 2 — module geography → GO (non-branching)

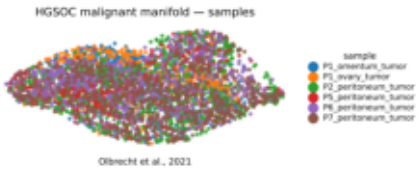
Gast/pSom co-vary (r=+0.44; +0.71 among non-cycling) and BOTH oppose Crest (pSom↔Crest r=−0.68); one developmental axis, Crest an inverse terminal gradient. Cell-cycle confound tested & rejected: anti-correlation survives excluding cycling cells; modules cc-free.

Gate 3 — hub cross-repression → NA

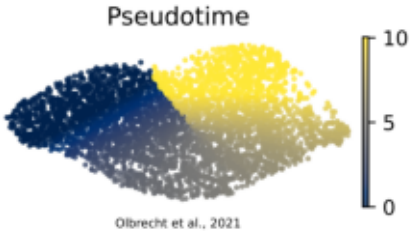
Branching-only. No bifurcation (curve: 2 tips / 0 forks, fold-free) — not tested. The non-branching case passes Gate 2 and reports Gate 3 as NA (distinct from a NO-GO where the pattern is absent).

STEP FIGURES

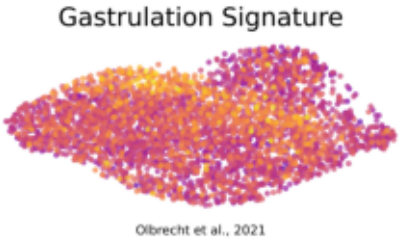
Manifold — samples



Pseudotime



Gastrulation



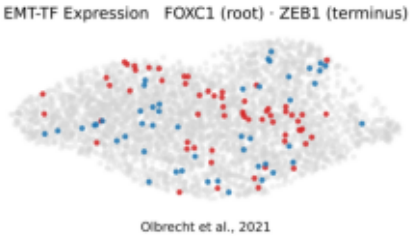
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

High-Grade Serous Tubo-Ovarian Carcinoma (Olbrecht et al., 2021), 10x — 3,936 malignant cells across 6 samples / 5 patients (≥155-cell threshold keeps 6/10 samples). Harmony integration on sample (θ=10), principal-curve trajectory (scFates) on the whole manifold.

Verdict.

Pattern REPRODUCES (non-branching). The embryonic programme recapitulates on a single continuous axis: a Gast/pSom-high origin resolving into an inverse, terminal Neural-Crest gradient (pSom↔Crest r=−0.68; along pseudotime Gast −0.33, pSom −0.06, Crest +0.40 terminal). Linear-pSom/Crest geometry (means Gast +0.029 / pSom +0.077 / Crest −0.020), not the Puram hub-and-spoke bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT: FOXC1 (root) → ZEB1 (terminus).

Parameter choices

Sample threshold	≥155 → 6/10 (gap 384→63, ratio 6.10; geom-mean cut)	**n_pcs**	20 (scree elbow; Kürten(20)/Puram(40) precedent)
Integration	Harmony θ=10 (iLISI 1.00→3.33, monotone)	**Island removal**	0 cells (UMAP radius-graph, single component)
Geometry	Non-branching curve (2 tips / 0 forks, fold-free)	**Root**	tip 0 = Gast/pSom-high; corr(x_lin,pt)=+0.84
Cell-cycle	whole manifold kept (pSom↔Crest survives non-cycling −0.63; cc-free)	**EMT-TF readout**	FOXC1 (root, ρ=−0.06) → ZEB1 (terminus, ρ=+0.11)